3-14-14

Physical Chemistry Cume

Quantum Mechanics

1) Multiple choice: (5 x 10 points each)

a) Which of the following functions is acceptable as a wavefunction in the range $-\infty < x < \infty$?

(i) e^{-x}

(ii) e^{x}

(iii) $\dfrac{1}{x^2+1}$

(iv) x

(v) $|x|$

b) If ψ is a normalized wavefunction, the mathematical expression $\int \psi^* \hat{H} \psi d\tau$ gives the value of

(i) The normalization constant of ψ

(ii) The most probable location of the particle

(iii) The average energy of the particle

(iv) The expectation value for the location of the particle

(v) The position of the particle where ψ is a maximum

c) If ψ is a normalized wavefunction for a stationary state of a hydrogen atom, the mathematical expression $\int \psi^* r \psi d\tau$ gives the value of

(i) The normalization constant of ψ

(ii) The most probable distance of the electron from the nucleus

(iii) The overall energy of the atom

(iv) The expectation value for the distance between the electron and the nucleus

(v) The distance from the nucleus where ψ is a maximum

d) If the Hamiltonian operator of a system is $\hat{H} = a\dfrac{\partial^2}{\partial x^2}$, an eigenfunction is $\psi = b\sin(kx)$, where a, b and k are constants. The energy for this state is given by

(i) $-abk^2$

(ii) $+abk^2$

(iii) $-ak^2$

(iv) $-bk^2$

(v) $-ak$

e) If a particle in a 1-D box of length a has an equal probability of being in the ground state and 1st excited state, a valid wavefunction for the system would be

(i) $\Psi(x) = \sqrt{\frac{2}{a}}\left(\sin\left(\frac{\pi x}{a}\right) + \sin\left(\frac{2\pi x}{a}\right)\right)$

(ii) $\Psi(x) = \sqrt{\frac{2}{a}}\left(\sin\left(\frac{\pi x}{a}\right) + 2\sin\left(\frac{2\pi x}{a}\right)\right)$

(iii) $\Psi(x) = \sqrt{\frac{1}{a}}\left(\sin\left(\frac{\pi x}{a}\right) + \sin\left(\frac{2\pi x}{a}\right)\right)$

(iv) $\Psi(x) = \sqrt{\frac{2}{a}}\sin\left(\frac{\pi x}{a}\right) + \sqrt{\frac{1}{a}}\sin\left(\frac{2\pi x}{a}\right)$

2) Show that, for a 1-D harmonic oscillator, a wavefunction in the form $\Psi(x) = e^{-gx^2}$ satisfies the Schrödinger equation, $\hat{H}\Psi = E\Psi$, when g is a certain constant. What is this constant, g, in terms of the mass and force constant of the oscillator.

HINT 1: The Hamiltonian operator for a 1-D harmonic oscillator is given by

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + \frac{1}{2}kx^2$$

Where m is the mass and k is the force constant of the oscillator

HINT 2: You may find the chain rule useful

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

HINT 3: You may find the product rule useful

$$\frac{d(u \cdot v)}{dx} = v\frac{du}{dx} + u\frac{dv}{dx}$$